Modeling and Forecasting Natural Gas Prices in India using Generalized Additive Models



Mid Semester Evaluation Report

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1 Introduction and Motivation

India's City Gas Distribution sector provides Compressed Natural Gas (CNG) to more than 6 million vehicles across over 100 cities. CNG prices are set locally by individual distribution companies and are affected by factors such as LNG spot prices, APM revisions, and local economic conditions. Conventional time series models such as ARIMA and OLS mainly capture linear relationships and may not fully represent changes caused by regulation, changes in demand, and other economic factors.

This study develops a Generalized Additive Model (GAM) to forecast CNG prices in India. GAMs allow the relationship between price and the explanatory variables to change across different levels of the variables through smooth functions. This makes the model suitable for studying relationships that are not strictly linear. The study tests whether a GAM with penalized cubic B splines and Fourier based seasonality can improve forecasting performance compared with an ARIMA model.

2 Data and Empirical Setup

2.1 Panel Dataset

Monthly CNG price data for six cities, Delhi, Mumbai, Noida, Ghaziabad, Ahmedabad, and Pune, were collected from distributor price revision notifications for the period from April 2011 to December 2025. This gives a total of 1,080 city month observations. The reported prices were converted into continuous monthly series using a Step Function Zero Order Hold and a Time Weighted Average Price calculation between consecutive price revisions. No synthetic price observations were added.

Macroeconomic variables were collected from MoSPI and RBI archives. These sources were combined into a monthly macro panel containing 177 observations. UN SNA Ratio Linking was used for base year splicing, while Denton Cholette temporal disaggregation was used to convert quarterly GDP data into monthly values. Descriptive statistics are presented in Table 1.

Table 1: Descriptive Statistics of Model Variables (2011–2025)

Variable	Mean	Std. Dev.	Min	Median	Max	Obs
CNG Price (INR/kg)	54.71	15.83	32.00	48.92	85.12	177
GDP Monthly (Cr INR)	1129554.78	281873.73	676790.10	1124258.59	1815282.22	177
CPI Index (Base 2012=100)	143.75	31.05	89.10	139.80	198.00	177
PFCE Income (Cr INR)	640011.58	163733.68	370875.15	623524.37	1046241.14	177

2.2 Multicollinearity Diagnostics and Gram Schmidt Orthogonalization

A major issue in the data was the high correlation between GDP and Private Final Consumption Expenditure (PFCE). Before estimation, the Variance Inflation Factor test gave a VIF of 46.24 for GDP and 45.38 for Income. To address this issue, modified Gram Schmidt orthogonalization was applied to the macroeconomic predictor matrix. After orthogonalization, the VIF for all predictors was 1.00. The resulting variables were then used in the GAM estimation. The model components are given in Table 2.

3 Model Specification and Results

3.1 Generalized Additive Model Formulation

The forecasting model is given by:

Table 2: Generalized Additive Model Structural Components and Multicollinearity Diagnostics

Component	Type	Parameters / Knots	Pre-VIF	Post-VIF
Intercept (β_0)	Constant	1	—	—
Secular Trend $P_d(t)$	Linear ($d = 1$)	1	—	—
Seasonal Basis $f_{\text{season}}(t)$	Fourier ($K = 2$)	4	—	—
Smoother g(GDP Monthly Cr)	Cubic B-Spline	df=5	46.24	1.00
Smoother g(CPI Index)	Cubic B-Spline	df=5	14.07	1.00
Smoother g(Income PFCE Cr)	Cubic B-Spline	df=5	45.38	1.00

$$\hat{Y}_t = \beta_0 + P_d(t) + f_{\text{season}}(t) + g_1(\widetilde{\text{GDP}}_t) + g_2(\text{CPI}_t) + g_3(\widetilde{\text{Income}}_t) + \varepsilon_t \tag{1}$$

where $P_d(t)$ represents the linear trend, $f_{\text{season}}(t)$ represents the seasonal component based on Fourier terms with $K = 2$ frequency pairs, and $g_j(\cdot)$ represents the penalized cubic B spline functions used for the orthogonalized macroeconomic variables. Each smoother was specified with 5 degrees of freedom. The model was estimated using `statsmodels.gam.api.GLMGam` with an identity link.

3.2 Walk Forward Cross Validation and Benchmark

An expanding window split was used for the analysis. The first 123 months, from April 2011 to June 2021, were used for training, while the following 54 months, from July 2021 to December 2025, were used for testing. An ARIMA model with order (1,1,2) was used as the benchmark and was selected through an AIC based grid search using the same training and testing period. The forecasting results are reported in Table 3.

Table 3: Out-of-Sample Forecasting Performance: GAM vs. ARIMA Benchmark

Model Specification	RMSE	MAE	MAPE	Superiority
ARIMA Baseline (Auto-Order)	28.0712	26.5008	33.42	Benchmark
Generalized Additive Model (GAM)	20.1040	18.7388	23.57	GAM (+28.38%)

In the test period, the GAM produced an RMSE of **20.10 INR/kg**, compared with 28.07 INR/kg for the ARIMA model. This corresponds to a **28.38% improvement** in RMSE. The result indicates that the smooth functions in the GAM are able to capture some of the changes in CNG prices that are not captured by the ARIMA model.

4 Conclusions and Next Steps

The main stages of the analysis have been completed. These include the construction of the CNG price dataset with 1,080 observations, preparation of the macroeconomic data, Gram Schmidt orthogonalization, GAM estimation, and comparison with the ARIMA benchmark. The GAM produced a 28.38% lower RMSE than the ARIMA model during the test period.

The next stage of the work will include adding city level fixed effects to account for differences between distributors, using Vahan vehicle registration data as a demand related variable, and preparing the complete research manuscript.

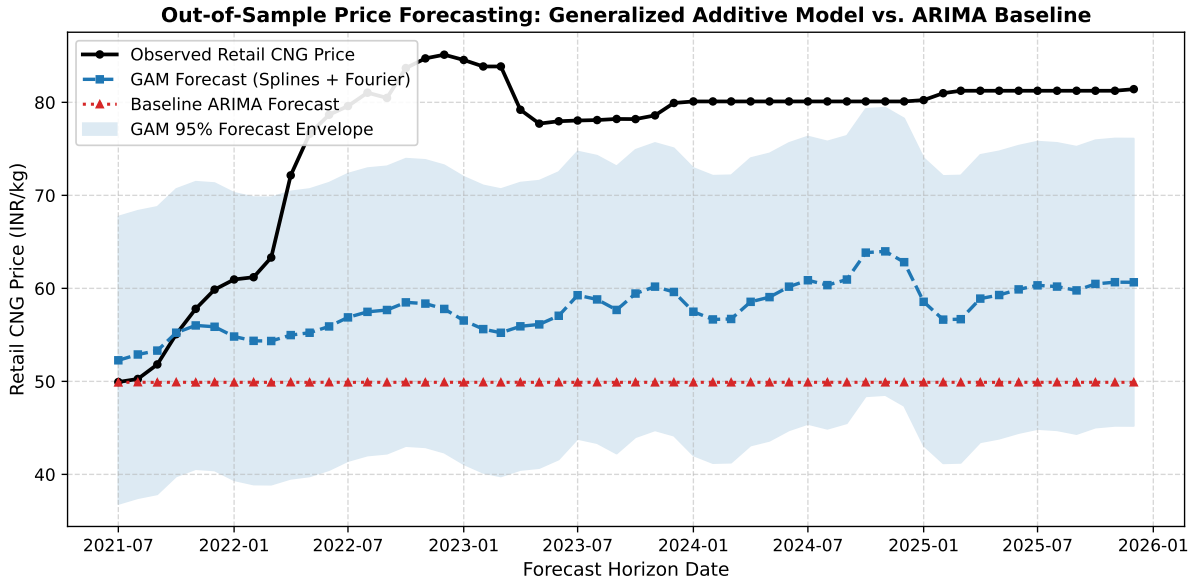


Figure 1: Comparison of GAM and ARIMA forecasts with actual CNG prices for the period from July 2021 to December 2025.

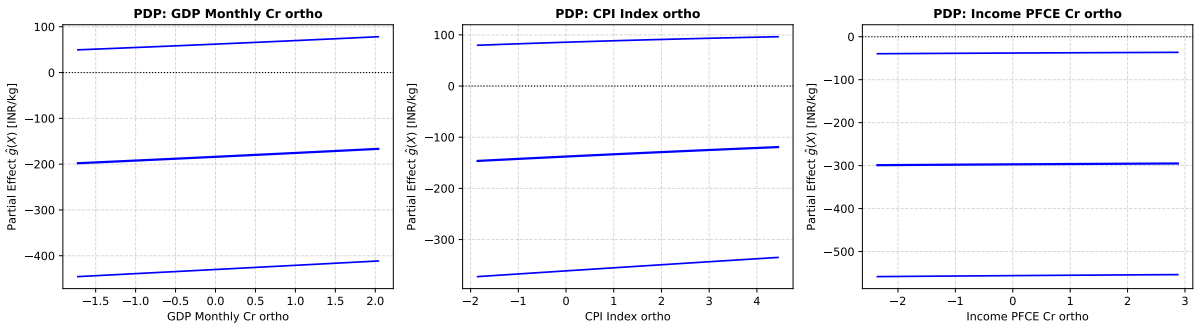


Figure 2: Partial dependence plots showing the relationship between GDP, CPI, Income, and forecast CNG price after orthogonalization.